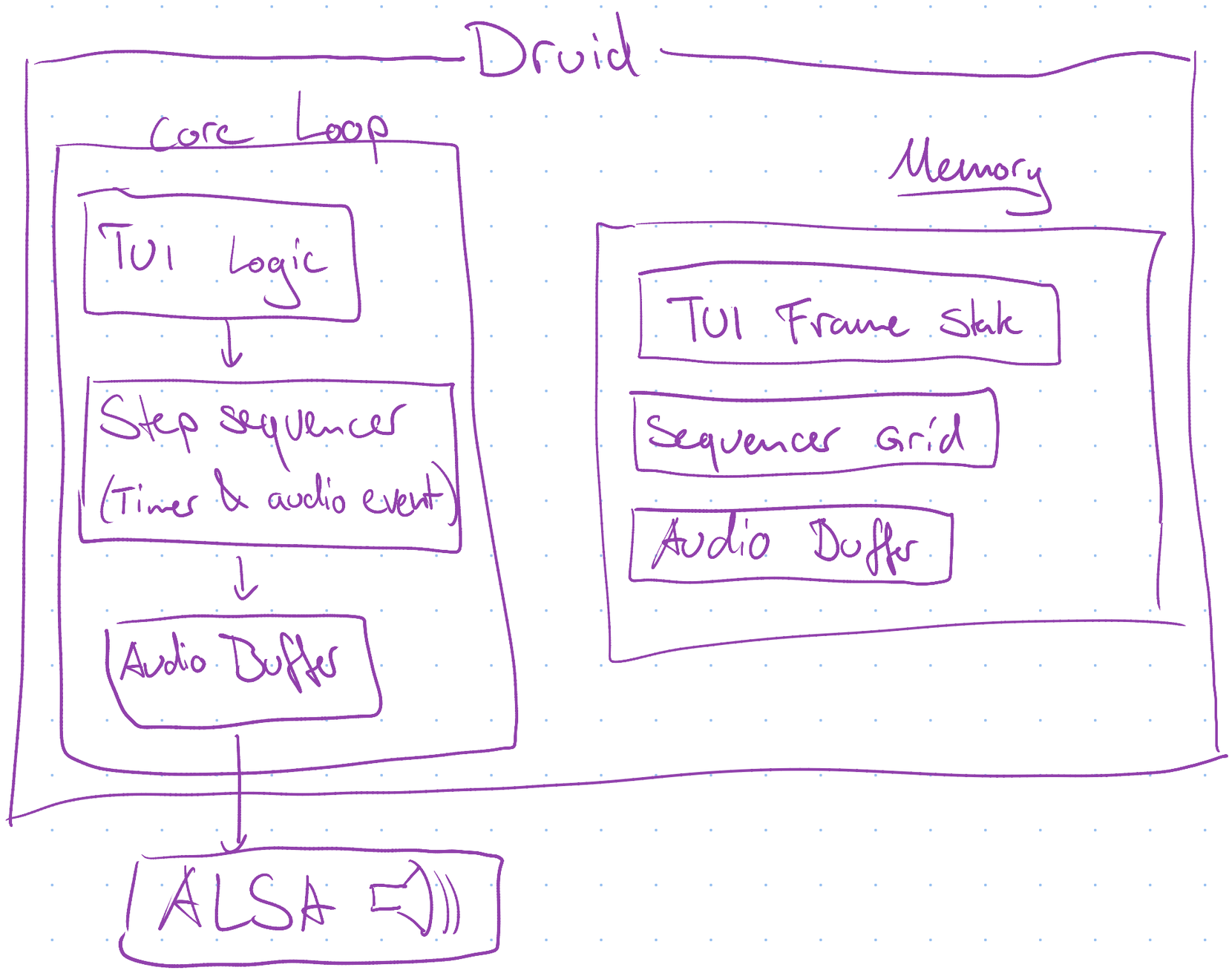
Druid

A Drum machine / Step sequencer for your Terminal, written in nasm Assembly.

- **Motivation:** This project relies on direct system calls, Memory handling of the Sequencer state & player location, as well as High-precision time/Audio handling.

- **Key features:** Syscalls will be used for printing to screen & setting the Terminal to raw mode. For the Audio, linking against ALSA & Timer Logic.

- **System Architecture:**



- Scope :

  Must haves:

  - Sequencer Grid & player logic
  - Audio out
  - Hard coded Audio samples

  Nice to haves:

  - adjustable BPM & swing
  - Wav files to play

- Potential challenges:

  - Audio output viability was researched with AI
    ↳ seems doable if it works as it says. If not then best to link against something that can handle Audio for me.
  - Seperate thread for Audio would be ideal to ensure smooth playback. → No clue how to do that
    ↳ worst case all in one thread with timing instabilities.